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''' Chinook Project: 500W Axial Flux Stator GeneratorTarget: 2026 KiCad PCB Design Contest Parameters
130mm OD, 12 Coils, 20mm Bore '''
import pcbnew
from pcbnew import *

def CreateChinookStator():
    board = pcbnew.GetBoard()
    num_coils = 12
    inner_r = 20.0
    outer_r = 65.0
    cx, cy = 150, 150
```

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    # 500W 48V spec: ~12A.

    # For 4oz copper, we need wide traces (~14mm total per phase path)
    trace_width = pcbnew.FromMM(2.0) # Individual turn width
    spacing = pcbnew.FromMM(0.5)

    for i in range(num_coils):
        angle_start = (i * 360 / num_coils)

        # Generate trapezoidal wedge coils
        print(f"Generating Coil {i} at {angle_start} degrees...")

        # (Simplified logic for contest submission script)
        # Actual implementation would use pcbnew.DRAWSEGMENT or pcbnew.TRACK

    pcbnew.Refresh()
```

```
if name == "main":
    CreateChinookStator()
```